

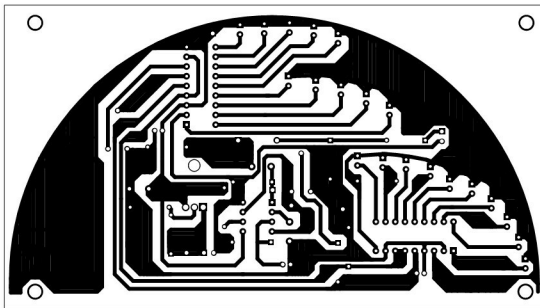


Fig. 2: An actual-size, single-side PCB for the RPM meter

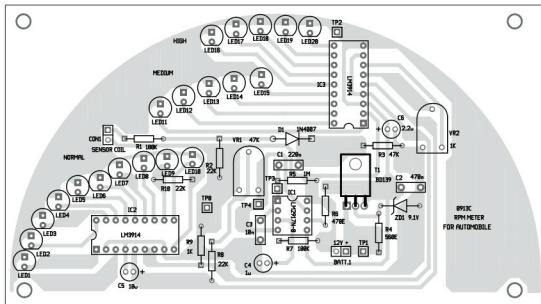


Fig. 3: Component layout for the PCB

this output voltage can be controlled by preset VR1. LM2917N-8 (IC1) is available in 8-pin and 14-pin versions; we have used 8-pin version here. The output of IC1 is fed at pins 5 of IC2 and IC3 through a filter formed by resistor R8 and capacitor C5. This filter removes any variations in the output and provides stable reading. IC2 and IC3 are connected to each other in cascade mode. Pin 9 of both these ICs are held at high voltage to select bar mode.

The number of LEDs glowing will

be directly proportional to the input voltage at pin 5, which, in turn, is dependent on the frequency of pulses. This implies that the number of LEDs glowing will be in accordance with the RPM of the vehicle.

The overall system can be calibrated as per one's liking (see last paragraph). We calibrated it such that each LED indicates about 1200 revolutions. LEDs should be of three different colours so that these can indicate normal (green), medium (blue) and high (red)

speed zones.

Power supply is taken from the vehicle's battery. Car battery's 12V DC is reduced to 9V so that the display's brightness remains the same, irrespective of whether the battery is less charged or over-charged.

Construction and testing

An actual-size, single-side PCB for the RPM meter is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable box. Use two-pin connectors on the PCB for connecting the sensor coil and the battery terminals.

To check the circuit for proper functioning, verify input supply at TP1 with respect to TP0. This supply is reduced to 9 V, which can be checked at TP2. Check the sensed pulses from the ignition coil at TP3 using an oscilloscope.

The voltage at TP4 will vary as per frequency of pulses.

To calibrate the system, as we did, use a function generator. Give 20Hz signal at TP3 and, using preset VR1, set its amplitude such that only LED1 glows. When you increase the frequency, you will see more LEDs glowing. Your system is now ready to use. Use preset VR2 to define the frequency range for each LED bar display. ●

The author is a regular contributor to EFW